3.3.1.3.2 Vertical Distribution of Seismic Forces

The vertical distribution of the pseudo lateral load shall be as specified in this section for all buildings except unreinforced masonry buildings for which the pseudo lateral loads shall be permitted to be distributed in accordance with Section 3.3.1.3.5. The lateral load F_x applied at any floor level x shall be determined in accordance with Equation (3-11) and Equation (3-12):

$$F_x = C_{vx} V \tag{3-11}$$

$$C_{vx} = \frac{w_x h_x^k}{\sum_{i=1}^{n} w_i h_i^k} \tag{3-12}$$

where:

C_{vx} = Vertical distribution factor

k = 2.0 for $T \geq 2.5$ seconds

= 1.0 for $T \leq 0.5$ seconds

Linear interpolation shall be used to calculate values of k for intermediate values of T

V = Pseudo lateral load from Equation (3-10)

w_i = Portion of the total building weight W located on or assigned to floor level i

w_x = Portion of the total building weight W located on or assigned to floor level x

h_i = Height (in ft) from the base to floor level i

h_x = Height (in ft) from the base to floor level x

3.3.1.3.3 Horizontal Distribution of Seismic Forces

The seismic forces at each floor level of the building calculated using Equation (3-11) shall be distributed according to the distribution of mass at that floor level.

3.3.1.3.4 Diaphragms

Diaphragms shall be designed to resist the combined effects of the inertial force, F_{px}, calculated in accordance with Equation (3-13), and horizontal forces resulting from offsets in or changes in the stiffness of the vertical seismic framing elements above and below the diaphragm. Forces resulting from offsets in or changes in the stiffness of the vertical seismic framing elements shall be taken as the forces due to the pseudo lateral load of Equation (3-10) without reduction, unless smaller forces are justified by a limit-state or other rational analysis, and shall be added directly to the diaphragm inertial forces.

$$F_{px} = \sum_{i=x}^{n} F_i \frac{w_x}{\sum_{i=x} w_i} \tag{3-13}$$

where:

F_{px} = Total diaphragm inertial force at level x

F_i = Lateral load applied at floor level i given by Equation (3-11)

w_i = Portion of the effective seismic weight W located on or assigned to floor level i

w_x = Portion of the effective seismic weight W located on or assigned to floor level x

The seismic load on each flexible diaphragm shall be distributed along the span of that diaphragm, proportional to its displaced shape.

Diaphragms receiving horizontal forces from discontinuous vertical elements shall be taken as force-controlled. Actions on other diaphragms shall be considered force- or deformation-controlled as specified for diaphragm components in Chapters 5 through 8.